

REPORT

Large-scale energetic and landscape factors of vegetation diversity

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Abstract

We show that at large scale the species number of vascular plants can be predicted to a major extent by climatically determined latent heat for evaporation and geometrical structure of landscape, described as an altitudinal difference. Application of the energy-equivalence rule across plant communities for transpiration per area unit, and use of fractal theory for the description of habitat occupation by vegetation results in a physically based species–energy relationship. Application of averaged global constants in the relationship results in a species–area equation with known parameters. Despite its simple form, this species–energy relationship generally reproduces global patterns of vegetation diversity at two scales – 10 000 and 100 000 km² – and is applicable for different regions across scales from 100 to million km². The proposed theory produces more robust results compared with the correlated-based approaches, which link plant diversity patterns with climate variables, and allows the inclusion of an evolutionary component. The final climate–richness equation for vascular plants has a simple and consistent analytical form and requires easily measurable variables.

Keywords

Energy equivalence rule, fractal dimension of landscapes, scaling of vegetation diversity, species number of vascular plants.

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INTRODUCTION

An explanation for the striking regional differences in the distribution of species of a variety of taxons across the Earth has long been a major task of biogeography and community ecology. The number of terrestrial and freshwater species within a sampling area of a given size has a visible latitudinal gradient, decreasing from the tropics to the poles in both hemispheres. This has been documented for morphologically different taxonomic groups (micro-organisms, trees, insects and primates) (Stevens 1989). A second robust biodiversity pattern, originally observed first in mammalian subspecies by Rapoport (1982), is characterized by a positive correlation between latitude and geographical range. Around 20 phenomenological and mechanistic explanations have been suggested to describe the existing latitudinal distribution of species (Brown & Lomolino 1998). One of the central points in debates on latitudinal gradients of species richness and geographical ranges is the role of scale dependence in mechanisms regulating biodiversity (Rosenzweig 1995). Latitudinal patterns for different taxa are disrupted significantly by

the influence of other factors such as high elevation or lack of precipitation. This indicates that the latitudinal dependence of species richness should be investigated mainly as a correlate of other environmental factors, like some form of total energy, available for species metabolism.

The type of available energy and the form of the relationship between this energy and the number of species in a given area are some of the major hot topics in recent biodiversity debates (Whittaker 1999; Gaston 2000). From polar to temperate regions there is evidence, based on observations, for a positive monotonic dependence of species richness with environmental energy, although the type of energy differs between taxonomic groups (Gaston 2000). The range of possible temporal variability in available energy appears to be a candidate for correlation with species richness in tropics and subtropics (Gaston 2000). Actual evapotranspiration, i.e. a measure of latent heat flux, was found to describe well the tree species diversity of North America and Great Britain (Currie & Paquin 1987); mean seasonal temperatures, i.e. measures of sensible heat, were best correlates with birds and butterfly species richness in the UK (Turner *et al.* 1987, 1988). The accumulative

measures of environmental energy such as net primary production can also provide a good description of species richness for certain taxa (for instance for tree species richness in temperate Europe, eastern North America and East Asia (Adams & Woodward 1989).

On the global scale correlation-based models relating species richness of vascular plants or their subsets with potential and actual evapotranspiration, annual temperature and water deficit were found to be globally very strong (e.g. O'Brien 1998; Francis & Currie 2003) at coarse spatial resolution. However, there is a need for more accurate models and theoretically sound explanations for the observed correlations between climate and vegetation diversity.

Here we suggest a species–energy relationship for vascular plant species, based on the rule of energy equivalence across plant communities in the form of equivalent transpiration per unit area, which is scale transparent and takes into account the seasonal distribution of available energy.

METHOD

Rule of energy equivalence across plant communities

The rule of energy equivalence can be derived from the more general model of energy optimization in a fractal network of branching transport tubes within an individual (Enquist *et al.* 1998, 1999a,b; Enquist & Niklas 2001). The metabolic rate of an organism is equal to its mass to the power 3/4 when the energy, dissipated during transportation of essential materials (water, blood, etc.), is minimized and the terminal tubes of the space-filling fractal transportation network do not vary with body size (West *et al.* 1997). The theory was successfully applied for the first time in plant ecology (Enquist *et al.* 1998, 1999a,b; Enquist & Niklas 2001), where previous allometric relationships were based on simple principles of Euclidian geometry. The geometric model predicts that the metabolic rate is proportional to the body mass to the power 2/3, i.e. the relationship between surface area (where heat is lost) and body mass (itself proportional to volume). Applying the resource distribution model through fractal networks to xylem transport of water and nutrients in vascular plants, a general model of plant vascular systems with the 3/4 law analogous to animal systems was suggested (Enquist *et al.* 1999b). This model was justified by using a 20-year time series, measuring 2283 trees from 45 species in Costa Rica (Enquist *et al.* 1999a). The data fits remarkably well the 3/4 power law relating the rate of gross primary production to body mass, when variation in wood density is taken into account. A 4/3 exponent for the intraspecific thinning law in plant populations was derived using this theory (Enquist *et al.* 1998). The new (fractal geometry based) thinning model

fitted well the observed data for 251 plant species ranging from *Lemma* to *Sequoia*, i.e. 12 orders of magnitude in plant size (Enquist *et al.* 1998).

Two important implications of the new allocation and thinning theory can be used in plant diversity studies. First, the rate of whole-plant xylem transport or transpiration is an appropriate index of plant metabolism, while the allometric exponents for gross photosynthesis, water and nutrient use must be equivalent due to stoichiometric constraints. Second, total energy use of plants in a given area is invariant of body size. Indeed, the total transpiration rate in a plant community of a given area Q_{tot} is the product of the rate per individual Q_{ind} and the population density N . Therefore:

$$Q_{\text{tot}} = N \cdot Q_{\text{ind}} \propto \alpha M^{-3/4} \beta M^{3/4} \propto \eta M^0, \quad (1)$$

where M is the above-ground plant biomass of an individual averaged across the community, α , β and η are constants, and hence Q_{tot} does not depend on M (Enquist *et al.* 1998; Enquist & Niklas 2001).

This strongly supports the rule of energy equivalence across plant communities on a given spatial scale: despite the fact that energy is utilized in a variety of growth form and life history strategies, all vascular plant species attain the same optimal use of energy in a community invariant of plant size.

Species–energy relationship for vascular plants

We can include the geometrical structure of landscape in the balance equation relating the available energy to the actual energy, used for transpiration in area A :

$$\gamma * E * A = Q * A_b, \quad (2)$$

where γ is the spatially averaged ratio between transpiration and total evapotranspiration, E is the latent heat per unit area available for evapotranspiration, A_b is total amount of transpiring area for vascular plants. Q is the constant rate of transpiration in energy units per unit transpiring area, according to the hypothesis of energy equivalence across plant communities.

Actual transpiration is distributed between species and therefore:

$$\gamma * E * A = Q * \sum_{i=1}^{N_{\text{sp}}(A)} A_b^i, \quad (3)$$

where $N_{\text{sp}}(A)$ is the number of terrestrial vascular plant species in area A and A_b^i is the total transpiring area for species i .

The total transpiring area for species i can be estimated using size–frequency distributions within an archipelago of self-similar transpiring islands (vegetation patches). Given that each part of the transpiring island can be described as a linear geometric reduction of the whole with the same

reduction ratios in all directions, the vegetation patches will be self-similar and their fragmentation, clumping and geometrical connectivity can be characterized by numerical characteristics used in the fractal theory (Mandelbrot 1983).

Self-similarity of vegetation patches for large scales was observed using aerial and remote sensing images at the community level for a variety of ecosystems such as Alaskan boreal forest (van Hees 1994), southern Texas savannah with the study area 12 million ha (Li 2000), mesophytic forest and cerrado in a 15 774-ha region of south-eastern Brazil (Jorge & Garcia 1997). Self-similarity for habitat occupation by vascular plants was also found at the species level for British grasses (Kunin 1998) and Alaskan trees (Lennon *et al.* 2002) from on-ground inventory data using a box-counting method (Mandelbrot 1983).

Hastings *et al.* (1992) suggested a power law relationship for cumulative number of self-similar islets within a distance r in a planar section as $N(r) = k * r^D$. The same power law was obtained by Voss (1988) when analysing raster images of natural objects. He shows that the number of square pixels $N(L)$ counted within irregularly shaped islands, which can be covered at an image by sampling square $L \times L$ window, can be estimated by the power law $N(L) = k * L^D$, where D is a fractal dimension of an image. On applying the power law to vegetation, the number of patches increases with the area of transpiring island a as:

$$N_{\text{patch}}^i(a) = k_i * \frac{1}{b_i} * a^{D_i/2}, \quad (4)$$

where k_i is a constant (maximum number of patches per unit area), $\frac{1}{b_i}$ is the reciprocal of relative landscape lacunarity (unitless value) (Milne 1992), D_i is the fractal dimension of vegetation in a landscape for the species i . The relative lacunarity $\frac{1}{b_i}$ is a prefactor, reflecting the influence of abiotic factors (edaphic, topographic, etc.) on density and connectivity of habitats, suitable for plants (Ritchie & Olff 1999).

Lennon *et al.* (2002) analysed fractal dimensions of species distribution for Alaskan trees and British grasses, i.e. vascular plants with different life forms and environmental conditions. They found that D_i may vary between 0.5 and 2. However, in both cases more than 70% of the total number of species possessed the range of fractal dimension between 1.3 and 1.9. The mean values of D_i for Alaskan trees and British grasses are close to each other (1.6 and 1.5 consequently). At first glance, we assume that $\frac{1}{b_i}$ and D_i depend mainly on geomorphological and long-term climatic parameters of the landscape and, therefore, relative lacunarity and the fractal dimension of vegetation at a landscape are constant across plant species, i.e. $\frac{1}{b_i} = \frac{1}{b}$ and $D_i = D$ for the entire set of species of the area A . The total transpiring area is calculated as the sum of areas of transpiring vegetation patches, where the number of patches can be

estimated using a power size–frequency distribution with $\frac{1}{b}$ and D constant across species:

$$\begin{aligned} \sum_{i=1}^{N_{\text{sp}}(A)} A_b^i &= \sum_{i=1}^{N_{\text{sp}}(A)} \bar{A}_i * N_{\text{patch}}^i(A) = \frac{1}{b} * A^{D/2} \\ &* \sum_{i=1}^{N_{\text{sp}}(A)} (\bar{A}_i * k_i) = \frac{1}{b} * A^{D/2} * \bar{\lambda} * N_{\text{sp}}(A), \end{aligned} \quad (5)$$

where $\bar{A}_i$ is the average area of a vegetation patch for species i , $N_{\text{patch}}^i(A)$ is number of transpiring patches of species i in the area A , $N_{\text{sp}}(A)$ is number of vascular plant species in area A , $\bar{\lambda} = \sum_{i=1}^{N_{\text{sp}}(A)} (\bar{A}_i * k_i) / N_{\text{sp}}(A)$ is the averaged constant, characterizing packing of plant vascular species in a landscape.

Combining eqns (3) and (5) provides an estimate of the number of vascular plant species in area A using general landscape characteristics and energy components:

$$N_{\text{sp}}(A) = \left(\frac{\gamma}{Q_{\text{tot}}^{\text{sp}}} \right) * E * b * A^{2-D/2}, \quad (6)$$

where $Q_{\text{tot}}^{\text{sp}} = Q * \bar{\lambda}$ is the average rate of transpiration for one species per area unit, constant in a landscape.

The maximum annual available latent heat per unit area can be obtained as a long-term averaged value:

$$\bar{E} = \sum_{j=1}^{N_{\text{year}}} \sum_{T_j > 0} \frac{\min(LH_j; L_j)}{N_{\text{year}}}, \quad (7)$$

where LH_j is the portion of radiation available for evaporation in month j with positive average temperature, L_j is the energy required to evaporate precipitation in month j , N_{year} is the length of the averaging period, which should be long enough to mask effects of interannual climate variability. LH_j can be calculated (in MJ m^{-2}) from monthly temperature, applying the dependence of surface temperature on the radiation balance as used in the energy-balance climate models (Ramanathan & Coakley 1978; Ramanathan *et al.* 1979; Balobaev 1991):

$$LH_j = \beta * (F + K * T_j) * t_{\text{month}}, \quad (8)$$

where T_j is the mean monthly temperature ($^{\circ}\text{C}$), $F = 49.6 \times 10^{-6} \text{ MW t m}^{-2}$ and $K = 2.1 \times 10^{-6} \text{ MW t (m}^2 \text{ }^{\circ}\text{C)}^{-1}$, $t_{\text{month}} = 2.592 \times 10^6 \text{ s}$ is the number of seconds in a month and β is equal to the global value $2/3$ (Baumgartner & Reichel 1975). $L_j = L * P_j$, where $L = 2.45 \text{ MJ kg}^{-1}$ is the latent heat of evaporation and P_j is the monthly precipitation in mm (equivalent to kg m^{-2}).

This approach allows consideration of seasonal distribution of available heat and water for plants, which is important for regions with summer (winter) dry climates, like India or continental Brazil.

Average global energy constants

The average global ratio between transpiration and total evapotranspiration was estimated as $\gamma = 0.52$ (Choudhury *et al.* 1998), which results in a land surface average transpiration of 284 mm year^{-1} or $Q = 696 \text{ MJ year}^{-1} \text{ m}^{-2}$ of latent heat, with a total annual evapotranspiration of 547 mm year^{-1} (Baumgartner & Reichel 1975; Choudhury *et al.* 1998).

A rough estimate of the annual average rate of transpiration for one species per square metre $Q_{\text{tot}}^{\text{sp}}$ can be calculated from the global average number of vascular plant species per unit area, using eqn (6) for species numbers of vascular plants. Indeed, when A is equal to a unit area; b is set to 1; γ , Q and E have average global values, the constant $\bar{\lambda}$ can be calculated from eqn (6) as the reciprocal of the global average number of vascular plant species per unit area. We take this number 4.5 species per square metre, which coincides with the average value for temperate broad-leaved forest (4.38–4.8 species per square metre) (Gleason 1922), i.e. for the zone with average global climatic conditions. Using this value we calculate an average of $Q_{\text{tot}}^{\text{sp}} = 154 \text{ MJ}/(\text{m}^2 \cdot \text{species})$.

Thus we obtain an average efficiency of energy used for species $v = \frac{\gamma}{Q_{\text{tot}}^{\text{sp}}} = 0.0036 \text{ species/MJ}$ and a species–energy relationship for vascular plants:

$$N_{\text{sp}}(A) = v * \sum_{j=1}^{N_{\text{year}}} \sum_{T_j > 0} \min[\beta * (F + K * T_j) * t_{\text{month}}; L * P_j] / N_{\text{year}} * b * A^{2-D/2}. \quad (9)$$

This relationship allows to estimate the number of species of vascular plants at different scales from the climate variables as long as values of the fractal dimension of vegetation fragmentation D and lacunarity of landscape b are known.

This simple species–energy relationship can be developed further to include the temperature dependence of metabolic rate for plant communities in extreme climate conditions. Under such conditions the available environmental energy in the left side of eqn 2 will be limited by the metabolic rate of the community and is best modelled using a negative exponent relationship with ambient temperature (Boltzman relationship, see Allen *et al.* 2002).

Average landscape parameters and species–area relationship

The lacunarity of the landscape depends on abiotic geographical features making climatic or physical barriers hindering plant species migration, like altitudinal differences, presence of water bodies and rock outcrops.

We limit our analysis to large-scale landscape features and, thus, take into account only the influence of altitudinal

difference on lacunarity within area A , making it in a linear form:

$$b = 1 + R * \Delta H(A), \quad (10)$$

where R is a constant and $\Delta H(A)$ is the mean altitudinal difference (in m). The effect of elevation upon the surface radiation balance can be described approximately by a similar expression with $R = 10^{-4} (\text{m}^{-1})$, as higher elevations have both more open viewsheds and the solar beam travels through less air mass (Fu & Rich 1999). We take this R as an initial value to describe lacunarity changes in relation to the mean altitudinal difference.

We use $D = 1.5$ as the global average value of fractal dimension of landscape fragmentation, which corresponds to the classic fractional Brownian motion model of stochastic habitat occupation by plant species (Hastings *et al.* 1982; Sugihara & May 1990). Similar values were obtained during an assessment of vegetation heterogeneity in New Zealand (Nikora *et al.* 1999). On the North Island, vegetation patches covering areas from 1 to 10 000 km^2 have $D = 1.42$ whereas the South Island has a corresponding fractal dimension of 1.4.

After these considerations, energy theory logically leads us to the species–area relationship (SAR) with coefficients, which can be directly estimated from long-term climate variables and a landscape heterogeneity:

$$N_{\text{sp}}(A) = C * [1 + R * \Delta H(A)] * A^{\bar{\lambda}}, \quad (11)$$

where

$$C = v * \sum_{j=1}^{N_{\text{year}}} \sum_{T_j > 0} \frac{\min[\beta * (F + K * T_j); L * P_j]}{N_{\text{year}}} \quad (12)$$

is the average potential number of vascular plants per m^2 in area A and

$$\bar{\lambda} = \frac{2-D}{2} \quad (13)$$

is the floristic diversity index (Malyshev 1975).

Setting $\Delta H(A)$ to zero in the species–energy relationship (eqn 9) results in the oldest and best-documented functional dependence between number of species and area in community ecology, the Arrhenius equation (Arrhenius 1920) with $\bar{\lambda} = \frac{2-D}{2} = 0.25$ corroborated using field data representative of a wide range of animal and plant species (Preston, 1962) and several theoretical models (MacArthur & E.O. 1967; Pielou 1975; Sugihara 1980). The floristic diversity index $\bar{\lambda}$ has been related both to landscape fragmentation and plant species habitat occupation strategies (eqn 5) and is scale dependent. Crawley & Harral (2001) have shown that $\bar{\lambda}$ varies at small ($0.1\text{--}10 \text{ m}^2$) and intermediate ($10\text{--}1\,000\,000 \text{ m}^2$) scales from 0.2 to 0.5 and then drops at larger scales (10^8 to 10^{12} m^2) to 0.2–0.3.

Malyshev (1975) suggested ranges of $\bar{z}$ between 0.15 (deserts and tundra) and 0.36 (tropics). Landscapes with high vegetation fragmentation, caused for example by climate extremes like deserts or tundra, will have values of $\bar{z}$ lower than 0.25, while landscapes with uniform areas, like tropical evergreen forests, will have values close to this theoretical value, following from the classical Brownian movement model of migration (e.g. $\bar{z}$ for the five 50-ha plots of tropical forest across the globe) (see Plotkin *et al.* 2000).

With the estimated global average values for constants it is possible to calculate the number of vascular plant species in area $\mathcal{A}$ (in m^2) when climate variables and altitudinal difference are known:

$$N_{\text{sp}}(\mathcal{A}) = 3.6 \times 10^{-3} \sum_{j=1}^{N_{\text{year}}} \sum_{T_j > 0} \frac{\min[(81. + 3.8 * T_j); 2.45 * P_j]}{N_{\text{year}}} * [1 + 10^{-4} * \Delta H(\mathcal{A})] * \mathcal{A}^{0.25}. \quad (14)$$

RESULTS

To assess the applicability of the species–energy relationship (eqn 14) for vascular plant species, based on average global values of fractal dimension of fragmentation and lacunarity of landscape as well as average efficiency of energy use for species, we took data on regional species abundance for a wide range of area size and geographical location, and maps of global vegetation diversity. Global consistency of the final climate–richness relationship in comparison with the correlation-based models (Francis & Currie 2003) was tested over the well-known Wallace's phytogeographical provinces (Brown & Lomolino 1998).

Global vegetation diversity

The published global maps of species number of vascular plants were assembled per sampling area of 10 000 km^2 (Barthlott *et al.* 1999) and per sampling area of 100 000 km^2 (Malyshev 1991) to investigate this energy–diversity relationship. The contour lines in both cases were digitized and transformed by interpolation onto $0.5^\circ \times 0.5^\circ$ longitude/latitude computerized grid cell maps of observed species number (Venevskaya & Venevsky 2002).

The maximum available latent heat in the species–energy relationship was calculated from monthly fields of mean temperature and precipitation during 1901–98 using the CRU05 $0.5^\circ \times 0.5^\circ$ longitude/latitude climate data, available at <http://www.cru.uea.ac.uk/cru/data/>. The altitudinal gradients for the sampling areas of 10 000 km^2 and 100 000 km^2 were obtained for each $0.5^\circ \times 0.5^\circ$ longi-

tude/latitude grid cell by averaging elevation data from the Digital Elevation Model of the World (<http://edcwww.cr.usgs.gov/landdaac/gtopo30/gtopo30.html>).

The correlation between the theoretical and observed number of vascular plant species for both global data sets are: $r^2 = 0.77$ for the sampling area of 10 000 km^2 and $r^2 = 0.76$ for 100 000 km^2 (see Fig. 1a,b).

A step regression optimization for theoretical coefficients v , R and D was carried out for the data set with numbers of vascular plant species per 10 000 km^2 . The correlation coefficient improved slightly ($r^2 = 0.78$) and the optimized coefficients are theoretical, except for R : $v = 0.0037$ sp./MJ, $D = 1.52$ and $R = 3.0 \times 10^{-4} \text{ m}^{-1}$ (against the theoretical value of $1.0 \times 10^{-4} \text{ m}^{-1}$). This probably indicates that the altitudinal gradient affects landscape lacunarity by not only variation in the radiation balance but also by additional factors, and, therefore, the optimized value of R was used in further calculations.

The absolute number of species and their relative spatial distribution are generally reproduced for both data sets (e.g. Figs 2 and 3). Desert and tundra areas have lowest species number, tropics the highest. However, in some regions with extremely sparse vegetation (Central Australia, Arctic tundra), the numbers of species are slightly overestimated. Most likely this is the consequence of significant deviation from the global average value of landscape patchiness D or limitation of available energy in the form of the Boltzman's relationship for metabolic rates in a community (Allen *et al.* 2002). High diversity regions of south-western Australia and Cape Province are poorly reproduced. Accounting for specific ecological and evolutionary features of the regions may explain the bias.

Regional vegetation diversity at different scales

The influence of scaling on model performance was estimated using the observed values for number of vascular plant species in the former Soviet Union (FSU) (see Malyshev 1994), mapped over four different sampling areas: 100, 1000, 10 000 and 100 000 km^2 . The correlation coefficient between observed and simulated values for species number does not change for the three scales (0.77–0.76) and has the lowest value for the largest sampling area, i.e. 100 000 km^2 (0.6).

The mapped data for the FSU may have shortcomings in the analysis of scale independence, because of the similarity in data processing (i.e. similar methods of spatial extrapolation) for the four different sampling resolutions. The large number of points in the global and regional validation of the species–energy relationship also influences the correlation coefficients.

Hence, we also correlated the computed values for species number of vascular plants with the observed data

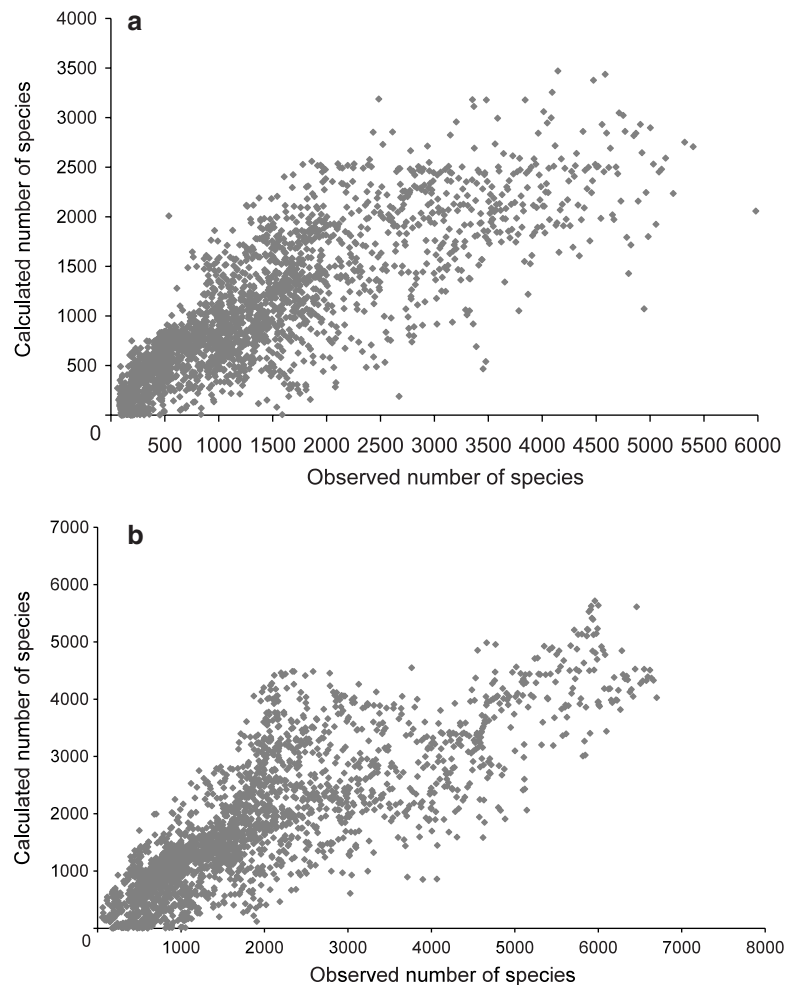


Figure 1 Observed against calculated species number of vascular plants across the globe (a) per 10 000 km² and (b) per 100 000 km².

from botanical surveys in different vegetation zones, from arctic tundra to the subtropics (see Tables 1 and 2). The data includes 59 observations from various authors for different area sizes, ranging from 100 to 4000 km² (Fig. 4a) in Europe, Asia, Australia, Africa and North America. Despite the variety of vegetation zones, geographical locations and area sizes, the observed and calculated number of vascular plant species is strongly correlated ($r^2 = 0.81$, $F = 241$, $P < 0.2 \times 10^{-22}$, slope = 1.03).

To investigate the upper scale limit of the species–energy relationship based on global constants, we analysed the survey data for countries and geographical provinces with significantly different species abundance, ranging from 0.8 to 16 million km² (Fig. 4b).

The correlation between the observed and theoretical number of vascular plant species per square metre averaged over the 16 regions ($r^2 = 0.83$, $F = 70$, $P < 0.0000076$, slope = 0.66) proves that the model can be applied for scales up to several million kilometres. However, one can see an overprediction in the regions

with sparse vegetation (like Alaska) and an underestimate in regions with dense vegetation (like Venezuela). The structural differences in landscapes for contrasting biomes, described by fractal dimension D , are more clearly seen for large scales, making estimates of species number more uncertain.

Testing of the climate–richness relationship for Wallace's provinces

The climate richness model for vascular plants, based on maximum available latent heat for evaporation and fractal characteristics of vegetation at the landscape, was compared against two recent correlation-based models (Francis & Currie 2003) for Wallace's provinces. The Francis & Currie (2003) models predict angiosperm richness using non-linear regression equations combining biologically significant climate variables, like annual temperature, potential evapotranspiration (PET) and water deficit (calculated as the difference between actual and PET).

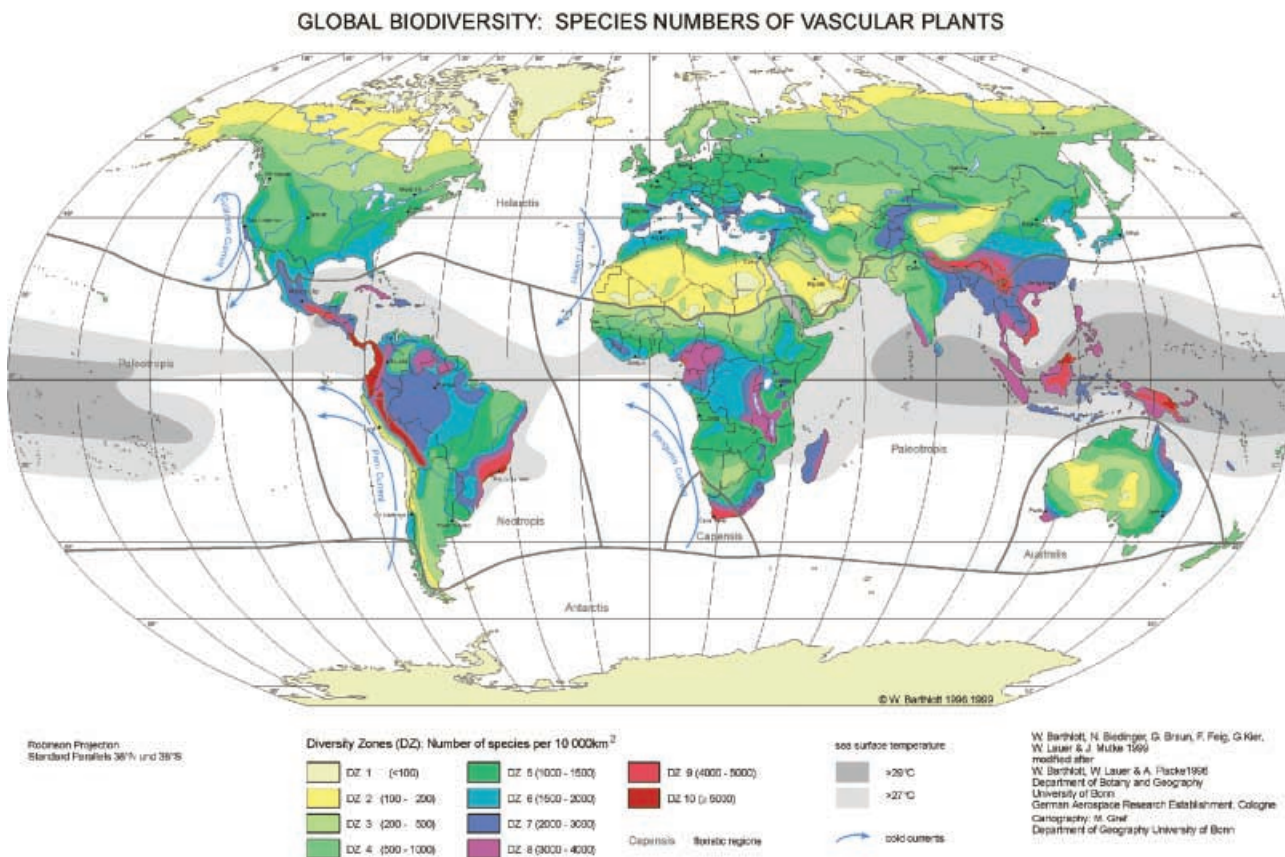


Figure 2 Observed vascular plant diversity pattern per 10 000 km² (<http://www.botanik.uni-bonn.de/system/biomaps.htm>).

The $0.5^\circ \times 0.5^\circ$ longitude/latitude monthly temperature and precipitation data and the data for altitudinal gradients were rescaled by cubic approximation onto a $2^\circ \times 2^\circ$ grid cell, the spatial resolution of the correlation-based models. The coarse resolution data set was divided into six regions, representing phytogeographical provinces similar to those used by Francis & Currie (2003) in their study. Species richness of vascular plants in a province, predicted by the species–energy relationship, was tested against the observed species numbers of vascular plants from the map of Barthlott *et al.* (1999) at the $2^\circ \times 2^\circ$ spatial resolution.

The resulting correlation coefficients demonstrate a higher consistency of the energy landscape model compared with the correlation-based ones (see Table 3). The species–energy relationship provides robust results for all six provinces (r^2 is between 0.65 and 0.78), while the correlation in the observed/predicted family richness for the polynomial regressions from climate variables varies significantly across Wallace's provinces, showing perfect fit for some provinces (e.g. $r^2 = 0.94$ for Nearctic) and relatively poor performance for others (e.g. the Oriental province, where $r^2 = 0.42$). The consistency of the energy landscape model

in different parts of the world demonstrates that the mechanisms of species and energy interactions are almost independent of the diverse evolutionary theory.

DISCUSSION

The final climate–richness equation for vascular plants (eqn 14) has a simple analytical form in comparison with formulations, suggested by the correlated-based approaches (see O'Brien 1993, 1998; Francis & Currie 2003). These approaches predict richness by extrapolation of non-linear regression equations (Francis & Currie 2003), which can switch input climate variables according to heuristic training rules (e.g. O'Brien 1998). While such models reproduce well the global diversity pattern for vascular plants at rather coarse spatial resolution ($>20\,000\text{ km}^2$) (Francis & Currie 2003), they have inevitable shortcomings, typical for regression-oriented approaches. The need to have the best fit leads to non-zero intercepts in regression equations, inferring negative values for species richness (e.g. O'Brien 1998), or positive values, when the PET is absent and vascular plants can not exist (e.g. Francis & Currie 2003). Spatial downscaling and imple-

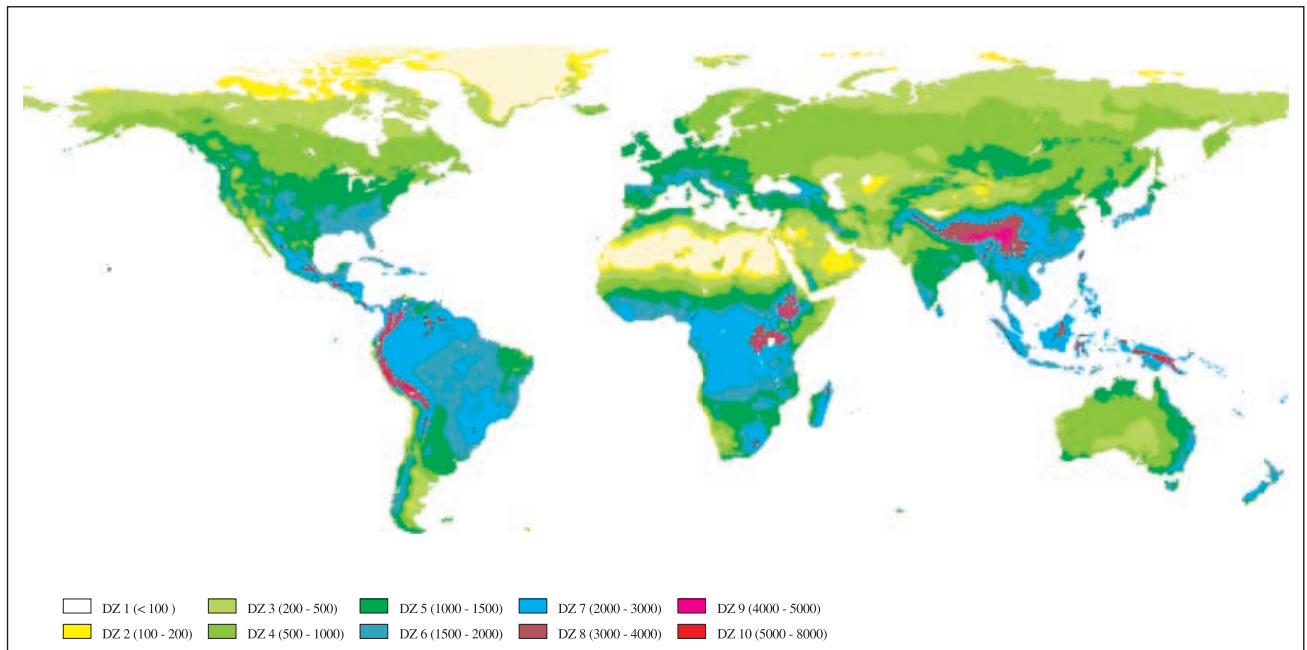


Figure 3 Simulated (using the species–energy relationship with the global average landscape and energetic parameters) vascular plant diversity pattern per 10 000 km². The ‘hot spot’ areas are obtained in Central America and in the Andes, on the Brazilian Coast and in Venezuela, in the western part of the Amazon basin, in Central and South Africa, in Madagascar, in south India, China, Indonesia and the eastern coast of Australia.

mentation of temporal dynamics also remain problematic for correlated-based approaches.

Despite its simple form, the proposed species–energy relationship with average global parameters reproduces the coarse scale global and regional patterns of vegetation diversity and works across scales from 100 to 10 000 000 km², keeping biological sense in the analytical formulation. It takes into account climate interannual variability and seasonality and can be used as a first estimate for species numbers of vascular plants in theoretical and practical studies. Stephenson (1998) analysed a variety of biologically meaningful climate correlates of vegetation distribution and concluded that the majority of them could not distinguish between climates with similar water and heat supplies, but different in their timing. The monthly timing of the maximum available latent heat of evaporation probably explains a better fit for the Oriental and Ethiopian phytogeographical provinces, achieved by the species–energy theory in comparison with global climate–richness regressions (Francis & Currie 2003).

The rule of energy equivalence across plant communities, which is the basis of the proposed large scale species–energy relationship, does not support the idea of regulation of species richness via energy-induced variations in the total number of individuals in a community (Brown 1984). Our relationship can be reformulated in a general form as: richness is equal to the average efficiency for energy use

across the species pool, multiplied by the maximum available external energy flux, seasonally distributed. This product is then scaled to the area considered, according to the habitat occupation pattern at a landscape. In this form the proposed species energy theory can be redesigned for any group of species, terrestrial or aquatic, because the largest external energy flux is usually known. The average efficiency for use of a certain type of energy for species within a community is the major intraspecific parameter, affecting richness in this case. For plant communities this parameter v can be influenced either by γ the ratio between transpiration and total evapotranspiration (efficiency of energy use of the entire community) or by packing of species at a landscape λ (species equitability in the community).

Efficiency of energy use for a community may have different character of change with community productivity, leading to different forms of the productivity richness relationship observed in nature, including positive, negative, hump-shaped or U-shaped (Mittelbach *et al.* 2001). Productivity in plant communities is determined to a major extent by evolutionary adaptation to local climate conditions and, most likely, we should expect a decrease in the efficiency of energy use after the productivity optimum is reached. This may explain the predominance of the peaked form in the productivity–diversity pattern for vascular plants (Mittelbach *et al.* 2001), amplifying at landscape (20–200 km) and local (<20 km) scales and damping at

Table 1 Observed and calculated species numbers of vascular plants (SNVP) for Europe (for observed values see Venevskaya & Venevsky 2002)

Zone	Locality	Longitude	Latitude	Area (km ²)	SNVPobs	SNVPcalc
Forest-tundra subzone	Nord Fugloy I, Norway	20	69	248	290	306
	Rastigassa, Lapland	26.2	69.8	265	292	322
	West Utsjoki, Finland	27.3	69.5	1075	310	409
Northern taiga subzone	Lapland Reserve, Russia	32.3	67.8	2784	523	682
	Khibin Mts, Kola Pen.	33.7	67.7	1800	429	564
Middle taiga subzone	Oulanka Nat. Park	29.2	66.4	107	429	291
Southern taiga subzone	Korpilahti, Finland	25.6	62	804	530	554
	Karku, Finland	22.1	61.3	190	503	394
	Sakyla, Finland	22.2	61	156	454	385
Subtaiga subzone	Nizhne-Svirsk Reserve	33.2	60.6	410	477	474
	Novgorod Province: Liubytino	33.4	58.8	700	538	598
	Latvia: mean of four areas	33.4	57.4	630	718	606
	Pskov Province: mean of 10 areas	29	57.3	707	700	640
	Novgorod Province: Kholm	31.2	57.2	750	587	611
	Pskov Province: Pushkinsky	28.9	57	750	604	622
	Latvia: mean of six areas	28.9	56.6	630	762	596
	Centralnolesnoy Reserve	32.9	56.5	213	546	458
	Pskov Province: Zhizhiza	31.4	56.3	750	607	623
	Lithuanian Natural Park	26.2	55.5	308	743	526
	Kurshskaya Kossa	21	55.1	160	630	502
	Beresinsky Reserve	28.4	54.7	760	768	663
	Naliboki, Belorussia	26.5	53.7	2400	820	909
	Belovezhs. Pucha	24	52.8	876	889	729
	Pripiatsky Reserve	28	52	603	740	665
Nemoral subzone	Shatskye Osera	23.8	51.6	710	825	702
	Polessky Reserve	28	51.5	201	602	499
	Kolbuszowa, Poland	21.8	50.2	2000	1001	952
	Strzyzow, Poland	21.6	49.8	1125	916	885
	Bieszczady Niskie	21.5	49.4	800	850	891
	Opolye Ukraine	24.5	49.3	4170	1298	1170
	Cherkasskiy Bor, Ukraine	31.3	49.4	417	796	593
	Olt Gorge, Romania	24.3	45.2	500	958	949
Forest-steppe subzone	Bucharest-Danube Plain	26.3	44.2	1597	1180	915
	Babadag Plateau, Romania	28.3	44.8	600	994	652
Genuine steppe subzone						

regional (200–4000 km) and continental scales (>4000 km) in accordance with scale variation (Crawley & Haral 2001) of the floristic diversity index Z in the species–area relationship (eqn 11).

Despite clear dominance of climate factors in global vegetation distribution, other environmental processes, like evolutionary histories, post-glacial dispersal and soil nutrient dynamics may influence the recent spatial pattern. Historical factors, which played a substantial role in shaping contemporary vegetation patterns, are not described by the simplified species–energy relationship (eqn 9) and may partly explain the difference between observed and simulated values for species richness.

Indeed, periodical climate changes, caused by changes in the Earth's orbit (Milankovitch oscillations), lead to

considerable changes in the size and pattern of vegetation distribution. These recurrent variations in species distributions, decrease gradual speciation, increase the proportion of species formed by 'abrupt' genetic mechanisms and select against specialization (Dynesius & Jansson 2000). Generalist species with a flat plateau in the productivity efficiency of energy use relationship are less prone to extinction, caused by high climate variability and are also not prone to gradual speciation. Thus, unevenness of species packing in a community increases and the overall richness decreases on the evolutionary timescale (see changes of parameter $\bar{\lambda}$ in the species–energy relationship in eqn 6). When periodical climate changes on the evolutionary timescales are relatively small, species are prone to gradual speciation, favouring low dispersability

Table 2 Observed and calculated species numbers of vascular plants (SNVP) for the sampling area 100 km² (Malyshev 1991)

Continent	Locality	Longitude	Latitude	SNVP obs	SNVP calc
Asia/Oceania	Putoran Plateau	95	68	262	210
	Chukotka	-175	65	267	194
	Igarka	86.24	67.29	304	195
	Novosibirsk	83.03	56.03	463	305
	Low Amur River	135	49	500	391
	Mountains of S. Siberia	90	50	663	395
	Maritime south	131.56	43.07	473	442
	S. steppe, Kazakhstan	72	41	413	497
	Repetek	63.13	38.34	151	147
	Borzhomi	43.23	41.51	1263	917
	Japan Islands	140	39	544	454
	Mount Jamizo	138	35	920	990
	Savanna, north India	85	25	720	567
	Deccan Peninsula	75	17	635	600
	Aden	44.5	13	232	202
Australia/New Zealand	New Zealand	168	-47	452	690
	South-east Australia	150	-36	512	670
Africa	Sahara	0	18	156	132
	Basin of the river Congo	15	-16	596	457
	South Africa	28	-30	1217	824
	Rhodesia	28	-20	664	480
North America	Devon Island	-90	75	102	70
	East Greenland	-45	61	152	221
	USA mainland	-100	40	500	466
	South-Eastern States	-80	35	750	742
	Alaska (Ogot.-Creek)	-150	66	297	231

and high specialization, which can be amplified by frequent disturbances and/or poor soil nutrient conditions. High equitability of species in such a community (i.e. low value of λ) increases community efficiency of energy use v and species richness. This can be seen for example in the Cape Floristic Region (CFR) of South Africa and south-western Australia, the regions poorly captured by the species–energy relationship with the global constants (eqn 9). The Quaternary climate was relatively stable in the CFR and high vascular plant diversity was produced recently by high speciation rates, resulting in many plants with narrow habitat specializations and small geographical ranges (Cowling *et al.* 1998). Similarly, shrublands and woodlands in south-western Australia persisted through the last glacial maximum (Dodson 2001) and the regional plant richness is higher than those of south-eastern Australia, which also experiences a Mediterranean climate, but where vegetation communities were assembled from refugial areas.

To investigate historical factors a next-generation model should include possible non-linear effects at the evolutionary timescale, when energy use is influenced by plant species domination/equitability in a community, regulated by associated changes in gene flow, migration rates, seed availability and vegetation patchiness.

Vegetation patchiness D can deviate from its global average values, especially in regions with intensive land use. Artificial inputs in agricultural ecosystems have significant impact upon the efficiency of energy use by species within communities with implications for species richness, which should be studied. Some plant species can be affected by rapid climate changes, resulting in competition inequities with possible loss of future diversity.

Despite evidence of self-similarity of vegetation patches at large scales for important representative ecosystems, like boreal and tropical forests (Jorge & Garcia 1997; Lennon *et al.* 2002), this is not a settled issue for the entire globe and all the terrestrial ecosystems. Most likely in real-world distributions of vascular plants many species in a community may have self-similar or close to self-similar distributions, but others may not. Conditions and environmental mechanisms providing self-similarity of vegetation patches at the species level across the globe should be identified and implemented in the next steps, while the assumption of fractal distributions for vascular plant species provides a good first approximation to estimate their number.

We obtained the classic power-law SAR assuming that individual species in a vegetation community have fractal distribution with the same dimension. Harte *et al.* (1999)

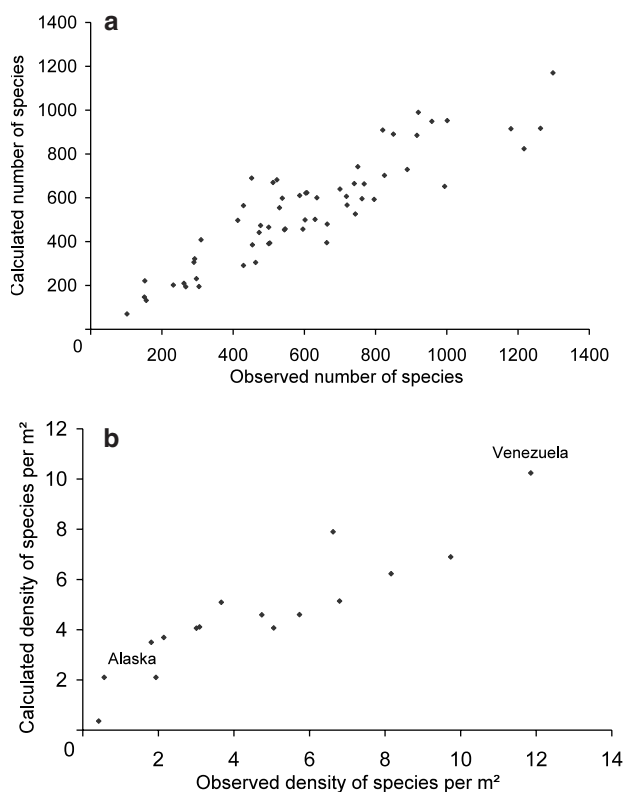


Figure 4 Observed against theoretical species numbers of vascular plants (a) for various areas from 100 to 4500 km² and (b) per square metre for large regions with variable areas from 0.84 to 16.48 million km² (observed values are collected in Heywood & Watson 1995).

Table 3 Performance of the three climate richness models for vascular plants within Wallace's biogeographical provinces

Province	Energy landscape model (r^2)	Temperature models (r^2)	PET models (r^2)
Nearctic	0.75	0.94	0.94
Neotropical	0.65	0.67	0.84
Palaearctic	0.77	0.79	0.89
Ethiopian	0.73	0.61	0.44
Oriental	0.71	0.43	0.42
Australian	0.78	0.66	0.78

Energy-landscape model: richness = maximum latent heat for evaporation $\times$ geometrical structure of landscape. Temperature models: richness = water deficit + temperature $\times$ water deficit (Francis & Currie 2003). PET models: richness = water deficit + PET + PET² (Francis & Currie 2003).

used other analytic methods to show that the constant fractal dimension in an assemblage of species leads to the classic SAR. However, our final SAR should be modified if a

type of probability distribution for fractal dimension between species and possible variation of fractal dimension across scales are known. Different observed types of SAR, e.g. sigmoid (Plotkin *et al.* 2000) or convex (Connor & McCoy 1979), may evolve when applying different distribution types for fractal dimension, as demonstrated and discussed by Lennon *et al.* (2002).

Physical and biological reasons for changes of lacunarity for species distributions at a relatively large scale should be described and implemented in the final species–energy relationship. The variation of lacunarity between species and regions should be combined and related with large scale variations in beta-diversity. Little is known about mechanisms of large-scale beta-diversity so far. Condit *et al.* (2002) identified striking differences in beta-diversity between Central Panama and western Amazonia and argued that this fact can not be explained by limited dispersal and speciation alone. Relatively low turnover of species in western Amazonia in comparison with Panama at large distances suggests that population densities of some species may be related to not yet identified processes (Condit *et al.* 2002). While geological variation between plant habitats is known to increase beta-diversity (Ruokolainen & Tuomisto 2002) other still unknown important factors influence the floras assembly in South America. Further research is needed to identify reasons of beta-diversity variation at large spatial scales, relating and implementing it into such a general parameter as the large-scale lacunarity of a region.

A next generation species–energy model based on separate parameter estimates for dominant vascular plant species in biomes (see eqns 3 and 5) may resolve these complicated issues. Such a model can provide a new insight into a process-based modelling study of global vegetation distribution (Kleidon & Mooney 2000).

Nevertheless, a substantial fraction of the variation in vegetation diversity can be explained by limitation in available environmental energy (maximum available latent heat) and by matrix landscape structure (altitudinal difference) and estimated using global average constants.

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